# DEPTH OF THE MODULE OF HOMOMORPHISMS INTO A MODULE OF POSITIVE DEPTH

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

*Editorial note.* Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

*Editorial note.* The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

*Allowed standard prerequisites (editorial).*

- Depth of a finite module over a Noetherian local ring and its behaviour under direct sums. used at proof lines 5395-5408. Supplies the quantity that is being bounded.
- A submodule of a module of depth at least one has depth at least one. used at proof lines 5400-5403. Gives the first part of the statement.
- Depth lemma for a short exact sequence of finite modules over a Noetherian local ring. the author cites Algebra, Lemma algebra-lemma-depth-in-ses at proof line 5406. Transfers the bound from the two outer terms to the kernel in the second part.

*Source reference.* Original lemma 0AV5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13673–13673. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 00LX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 803–803. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $R$ be a Noetherian local ring. Let $M$, $N$ be finite $R$-modules.*

(1) *If $N$ has depth $\geq 1$, then $\mathrm{Hom}_R(M, N)$ has depth $\geq 1$.*
(2) *If $N$ has depth $\geq 2$, then $\mathrm{Hom}_R(M, N)$ has depth $\geq 2$.*

*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* Choose a presentation $R^{\oplus m} \to R^{\oplus n} \to M \to 0$. Dualizing we get an exact sequence

$$0 \to \mathrm{Hom}_R(M, N) \to N^{\oplus n} \to N' \to 0$$

with $N' = \mathrm{Im}(N^{\oplus n} \to N^{\oplus m})$. A submodule of a module with depth $\geq 1$ has depth $\geq 1$; this follows immediately from the definition. Thus part (1) is clear. For (2) note that here the assumption and the previous remark implies $N'$ has depth $\geq 1$. The module $N^{\oplus n}$ has depth $\geq 2$. From Algebra, Lemma 00LX we conclude $\mathrm{Hom}_R(M, N)$ has depth $\geq 2$. $\square$